

AERODYNAMICS & WEATHER PERFORMANCE

Airfoil Efficiency in Heavy Rain

Computational and experimental analysis of the NACA 0012 airfoil operating at low Reynolds numbers for wind turbine applications.

BY SUMIT RAM (AE25MO62)

| CORE OBJECTIVES

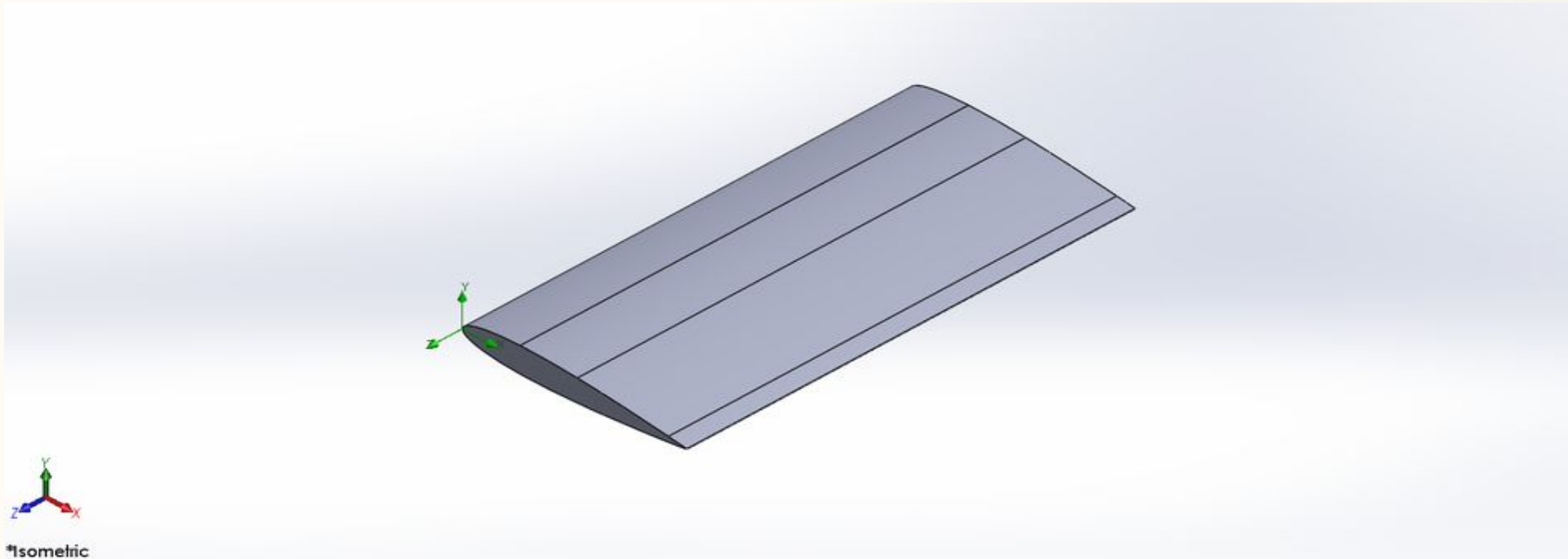
Performance Degradation

Quantifying the reduction in the Lift-to-Drag ratio when encountering varying Liquid Water Content (LWC).

Flow Visualization

Investigating the "Ejecta Fog" phenomenon and droplet breakup near high-pressure gradients using OpenFOAM.

THE AIRFOIL: NACA 0012



Geometry: Symmetrical profile, no camber, 12% thickness-to-chord ratio.

Relevance: A fundamental baseline used widely in wind turbine blades and aircraft vertical stabilizers due to its predictable behavior.

EXPERIMENTAL SETUP

Wind Tunnel

Open-circuit Plint TE44 tunnel.

Rain Simulation

Upstream nozzles injecting droplets at controlled flow rates.

Conditions

Liquid Water Content (LWC) ranging from 20 to 75 g/m³.

Instrumentation

3-component balance measuring Lift, Drag, and Moment.

| CFD TOOLCHAIN



C-Grid Mesh

Developed a structured C-type grid with high refinement in the boundary layer region.



OpenFOAM

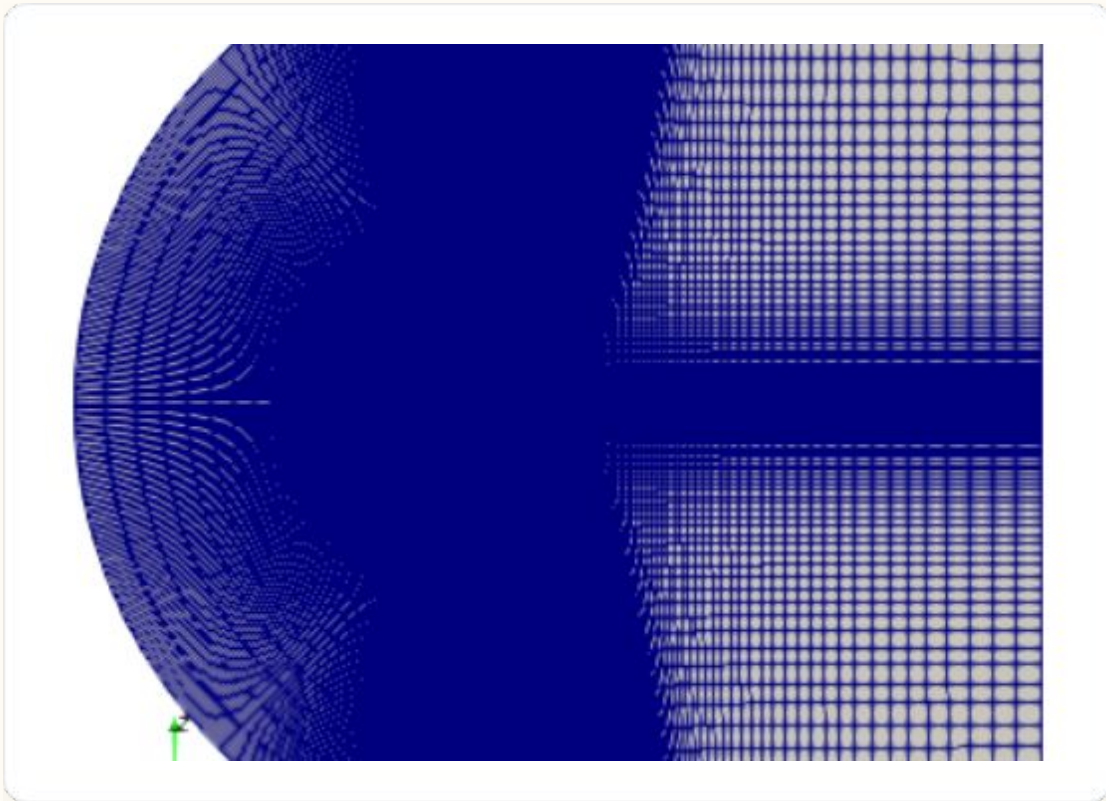
Utilized simpleFoam and icoUncoupledKinematicParcelFoam for one-way DPM coupling.



Python

Automated parametric studies across angles of attack and processed lift/drag data coefficients.

DOMAIN & MESH TOPOLOGY

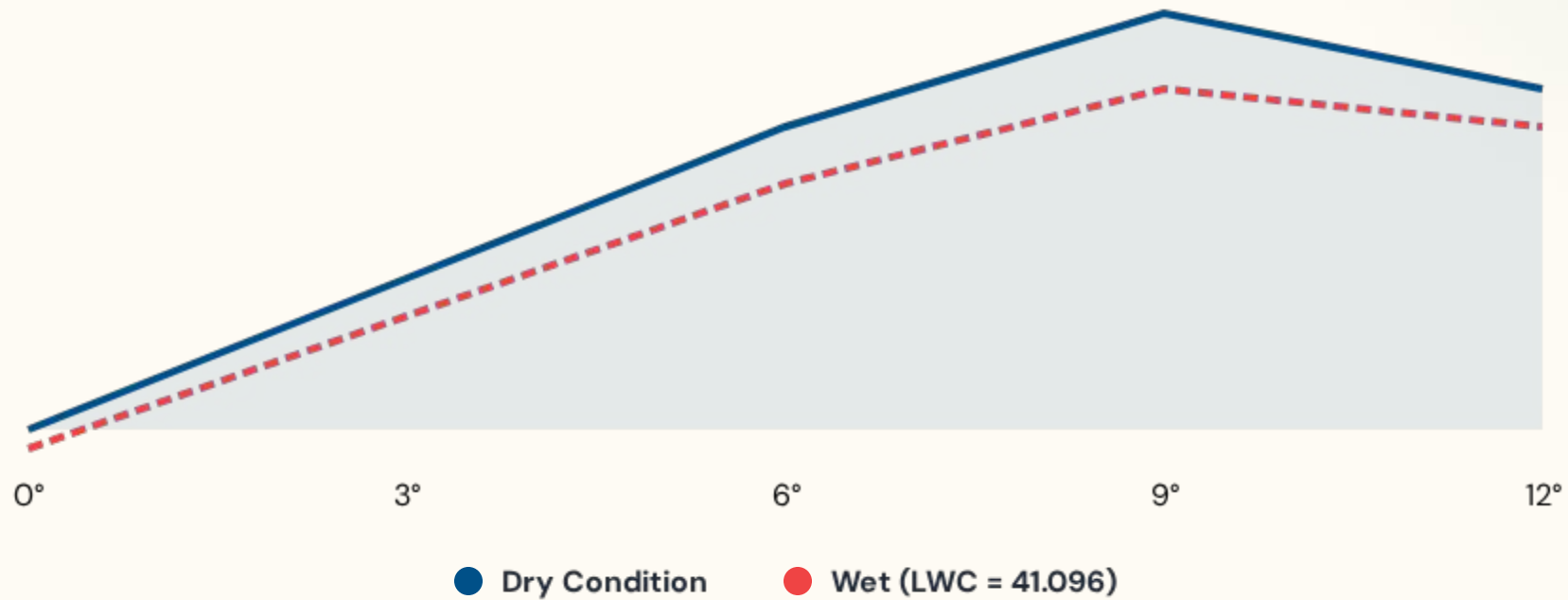


C-Type Grid Generation

A 2D continuous fluid domain was constructed using OpenFOAM's blockMesh utility.

- ▶ **Topology:** A C-type grid topology was utilized to establish a grid-independent solution..
- ▶ **Boundary Layer Resolution:** High cellular density near the airfoil surface to capture the boundary layer and "ejecta fog" formation.
- ▶ **Domain Scaling:** Outer boundaries set to ~20 chord lengths to prevent numerical reflection artifacts..
- ▶ **Droplet Injection:** Rainwater discrete phase released from a stable location 3 chord lengths upstream..

LIFT PERFORMANCE TRENDS



Observations show a distinct downward translation of the lift curve due to water film formation.

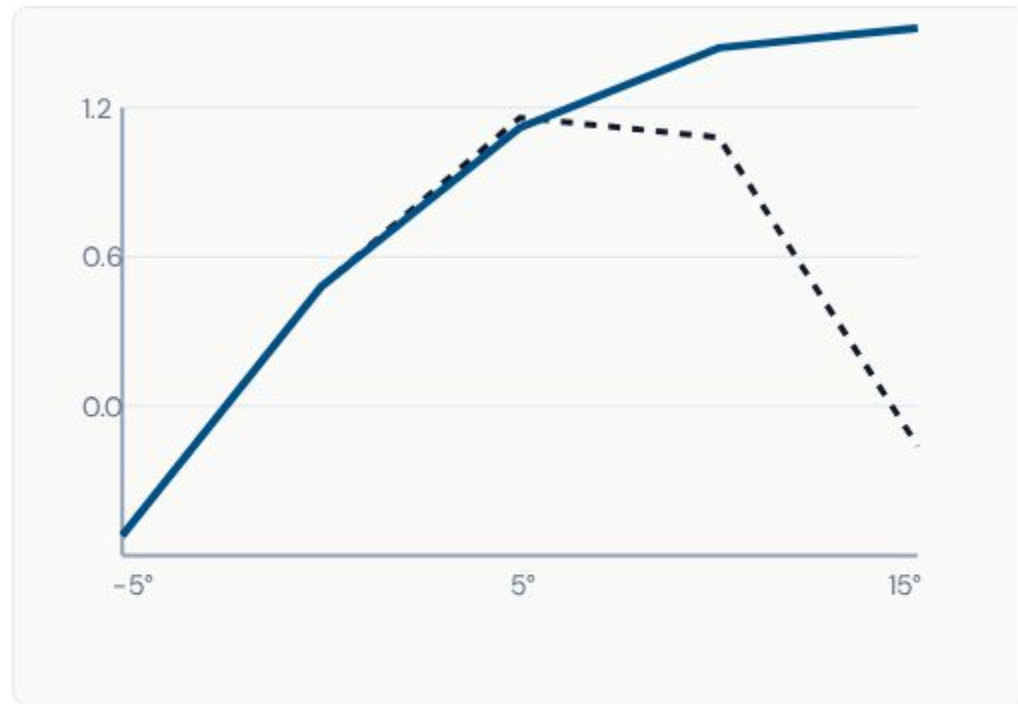
AERODYNAMIC COMPARISON

Metric (at AoA = 9°)	Dry Condition	Wet (LWC 75.491)	Impact %
Lift Coefficient (Cl)	1.12	0.74	-33.9%
Drag Coefficient (Cd)	0.03	0.08	+166%
Stall Angle	~10°	~12°	Delayed

1. BASELINE VALIDATION (DRY CONDITIONS)

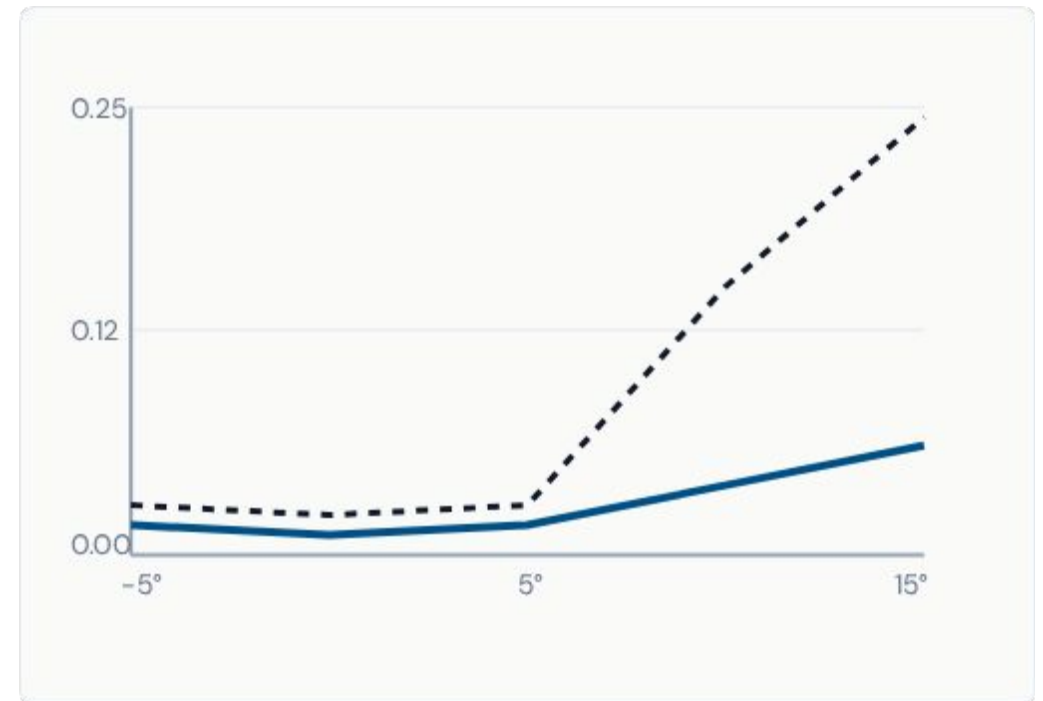
Comparing our CFD baseline against existing experimental data (Sheldahl and Klimas) before introducing multiphase rain..

Lift Coefficient (C_l) vs. Angle of Attack (α)



— Experimental (Sheldahl) — CFD Computational

Drag Coefficient (C_d) vs. Angle of Attack (α)

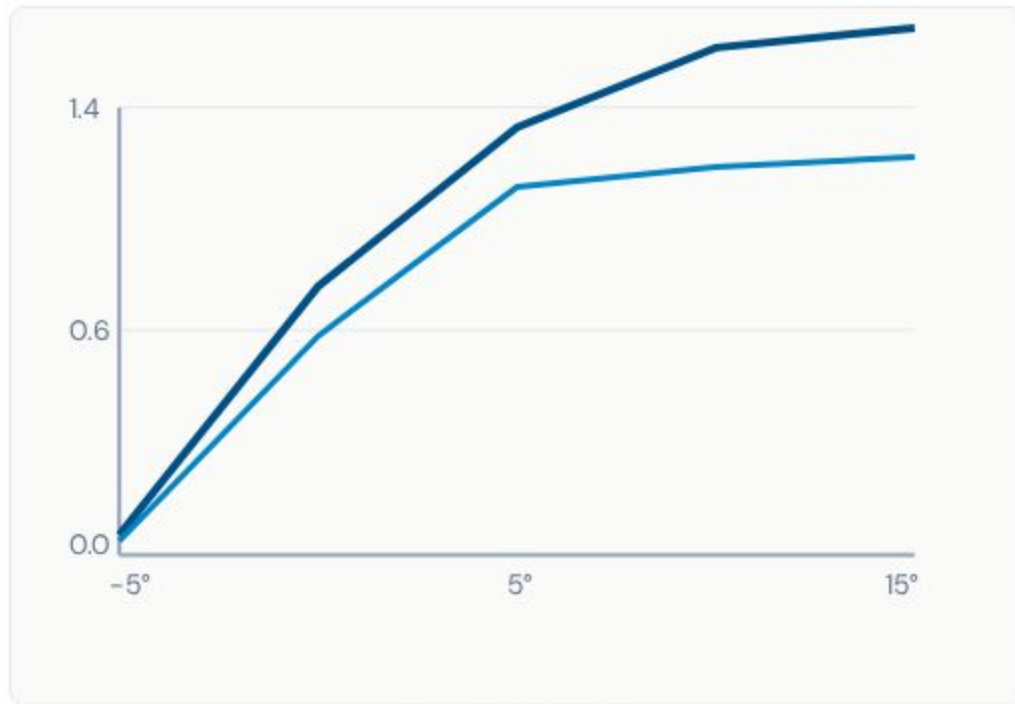


— Experimental (Sheldahl) — CFD Computational

3. LIFT DEGRADATION (HIGH LWC)

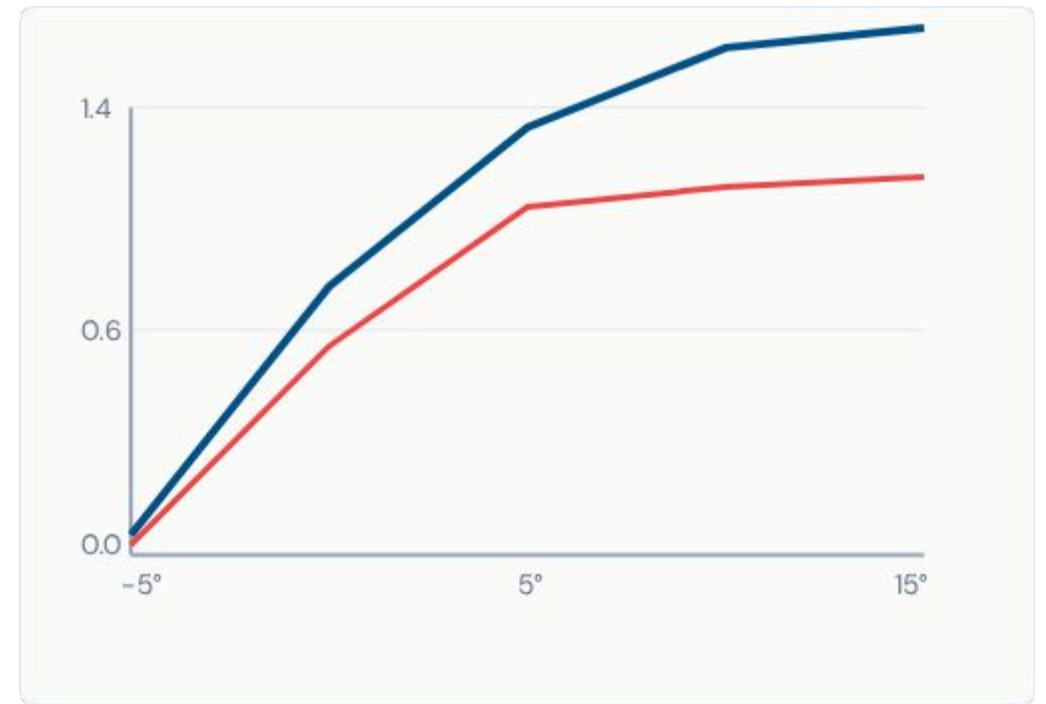
Heavy thunderstorm conditions cause severe water film formation and early boundary layer tripping.

LWC = 41.096 g/m³ (1.9mm droplets)



Maximum 31.4% reduction at 3° Angle of Attack

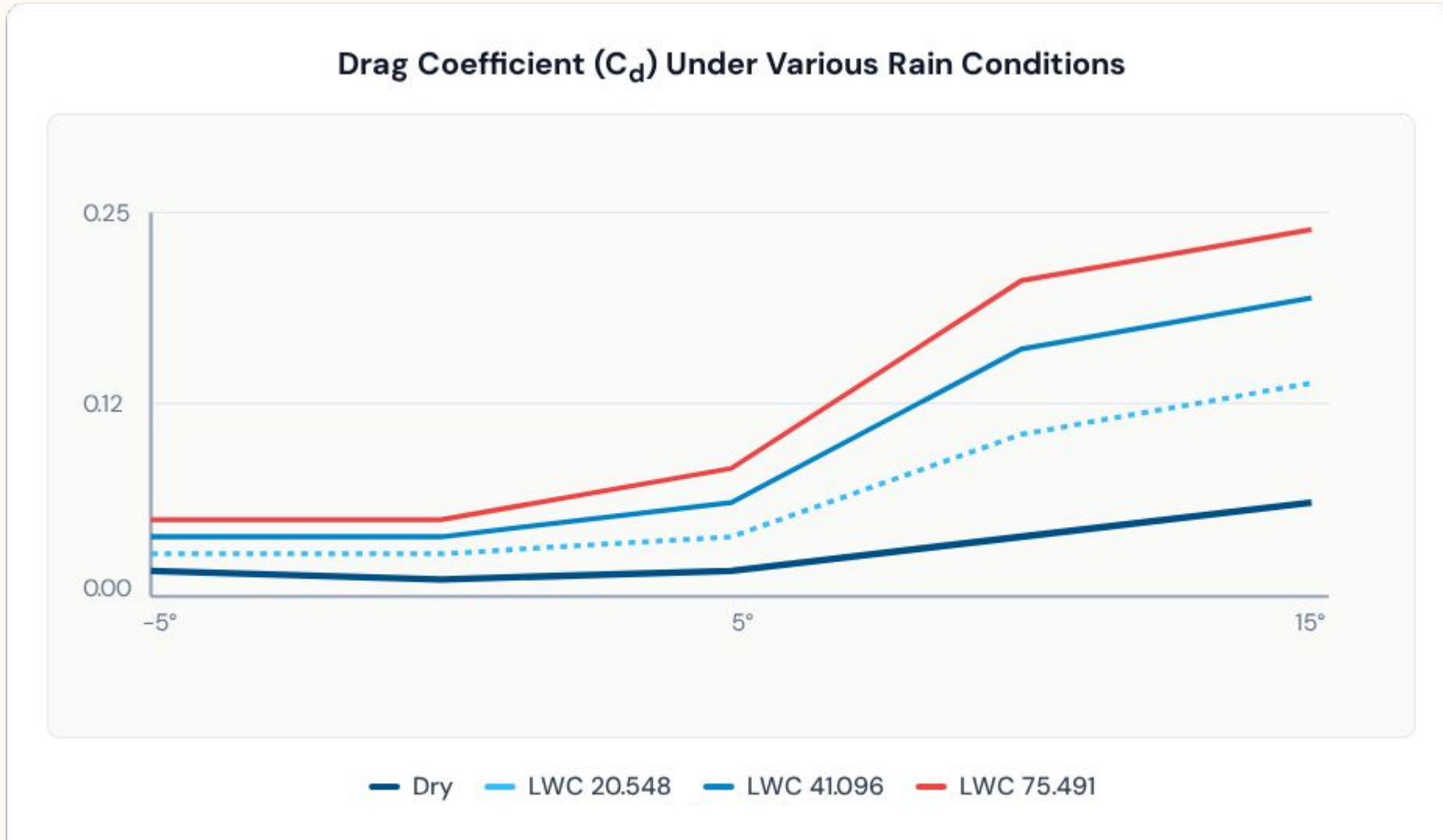
LWC = 75.491 g/m³ (1.2mm droplets)



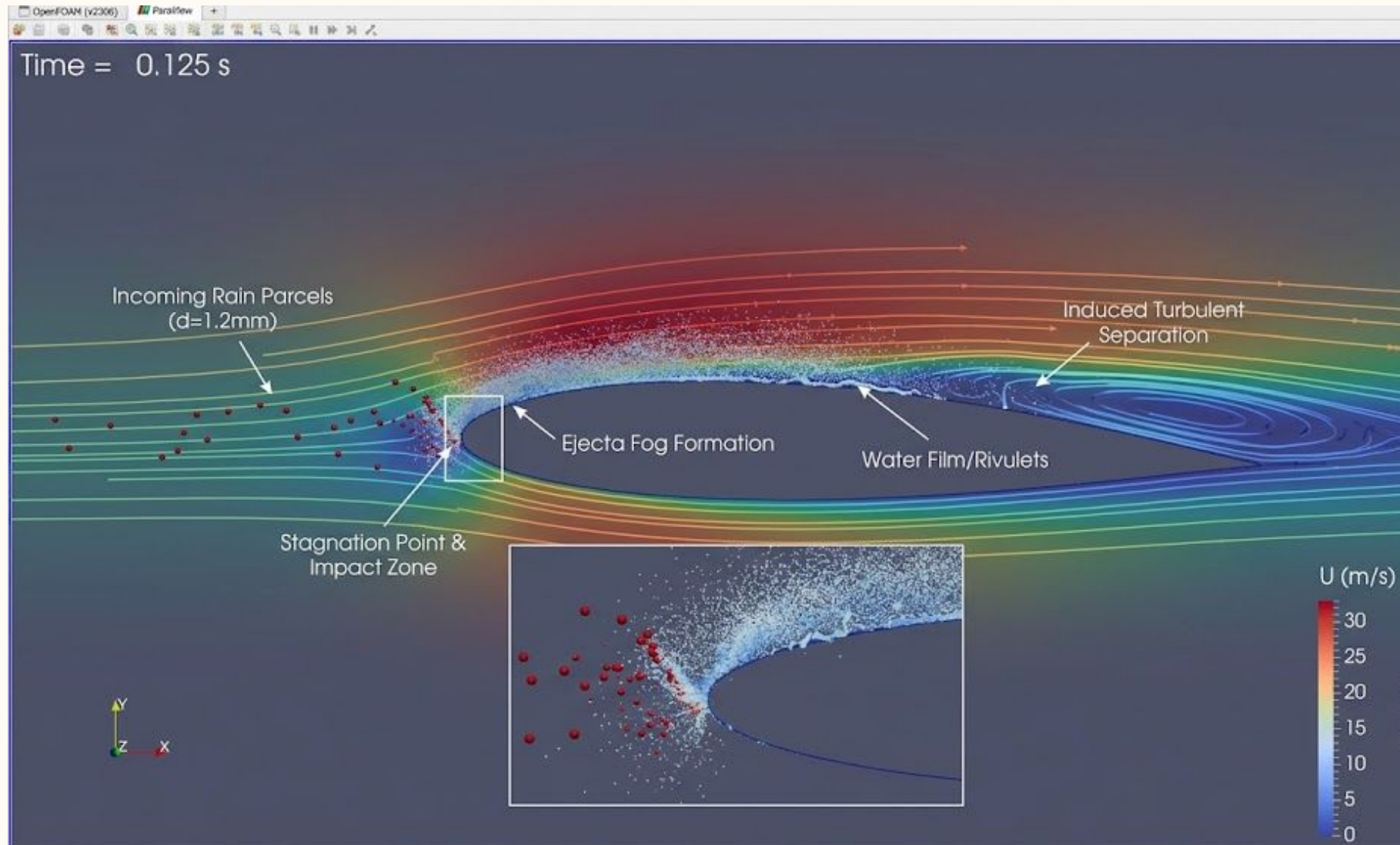
Maximum 33.2% reduction at 9° Angle of Attack

4. AGGREGATE DRAG PENALTY

The mixture of air and rain droplets leads to larger skin frictional drag across all conditions..



Visualisation of Ejecta fog on NACA 0012 Airfoil in openfoam



AERODYNAMIC PERFORMANCE SYNTHESIS

Lift Degradation

The primary effect of rain is a downward translation of the lift coefficient curve..

- ▶ Reduction increases with angle of attack until stall.
- ▶ Max penalty observed: **33.2% lift reduction** at 9° AoA for severe rainfall (LWC = 75.491 g/m³)..

Drag & Stall Behavior

Overall aerodynamic efficiency degrades dramatically in wet conditions

- ▶ **Drag Increase:** The air-water mixture leads to larger skin frictional drag and higher total drag..
- ▶ **Stall Delay:** For all simulated rain rates, the onset of aerodynamic stall was delayed to higher angles of attack.

Conclusion: Heavy rain acts as severe dynamic surface roughness, radically penalizing the Lift-to-Drag ratio of the NACA 0012 airfoil.

Machine Specification

AMD Ryzen 7 7445HS,

RTX 3050-4GB

8GB DDR5

Thank You

Questions & Discussion

Engineering Lab Project: NACA 0012 Weather Analysis